实验二. 小车基本操控

实验目的

1）了解 ROS2 中节点(node)、话题(topic)和数据(messages)概念
2）掌握 ROS2 基本操作指令

实验过程

下面指令，同组同学不能同时重复输入，但可以分工操作。

1）SSH 连接成功后，输入以下指令，启动机器人底盘驱动节点

ros2 launch originbot_bringup originbot.launch.py

注：启动成功后，可以听到机器人控制器的蜂鸣器短鸣 0.5 秒。

```
Last login: Sun Jan 12 07:58:19 2025
root@ubuntu:~# ros2 launch originbot_bringup originbot.launch.py
[INFO] [launch]: All log files can be found below /root/.ros/log/2025-01-12-08-00-27-935080-ubuntu-3664
[INFO] [launch]: Default logging verbosity is set to INFO
config_file_path is  /opt/tros/humble/lib/mipi_cam/config/F37_calibration.yaml
Hobot shm pkg enables zero-copy with fastrtps profiles file: /opt/tros/humble/lib/hobot_shm/config/shm_fas
Hobot shm pkg sets RMW_FASTRTPS_USE_QOS_FROM_XML: 1
[INFO] [originbot_base-1]: process started with pid [3665]
[INFO] [static_transform_publisher-2]: process started with pid [3667]
[INFO] [static_transform_publisher-3]: process started with pid [3669]
[static_transform_publisher-2] [INFO] [1736640030.047838180] [static_transform_publisher_Wy8Pz5bhaHtiNGvk]
orm
[static_transform_publisher-2] translation: ('0.000000', '0.000000', '0.053250')
[static_transform_publisher-2] rotation: ('0.000000', '0.000000', '0.000000', '1.000000')
[static_transform_publisher-2] from 'base_footprint' to 'base_link'
[static_transform_publisher-3] [INFO] [1736640030.047838097] [static_transform_publisher_z9AcPdPDe6O79YzC]
orm
[static_transform_publisher-3] translation: ('0.000000', '0.000000', '0.000000')
[static_transform_publisher-3] rotation: ('0.000000', '0.000000', '0.000000', '1.000000')
[static_transform_publisher-3] from 'base_link' to 'imu_link'
[originbot_base-1] [INFO] [1736640030.112458012] [originbot_base]: originbot serial port opened
[originbot_base-1] [INFO] [1736640030.647808094] [originbot_base]: IMU calibration ok.
[originbot_base-1] [INFO] [1736640031.148206853] [originbot_base]: OriginBot Start, enjoy it.
```

2）再启动一个 SSH 远程连接的终端，运行如下指令，启动键盘控制节点：

ros2 run teleop_twist_keyboard teleop_twist_keyboard

```
Last login: Sun Jan 12 07:59:34 2025 from 192.168.31.107
root@ubuntu:~# ros2 run teleop_twist_keyboard teleop_twist_keyboard

This node takes keypresses from the keyboard and publishes them
as Twist/TwistStamped messages. It works best with a US keyboard layout.
---------------------------
Moving around:
   u    i    o
   j    k    l
   m    ,    .

For Holonomic mode (strafing), hold down the shift key:
---------------------------
   U    I    O
   J    K    L
   M    <    >

t : up (+z)
b : down (-z)

anything else : stop

q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%

CTRL-C to quit

currently:      speed 0.5       turn 1.0
```

建议按 z 键，把车速 speed 降低到 0.1 以下，通过按 u、i、o 等键，控制小车运动。

3）查看节点(node)

启动第三个 SSH 远程连接的终端，输入

ros2 node list

```
root@ubuntu:~# ros2 node list
/originbot_base
/static_transform_publisher_DBKinOxjawqfEuwv
/static_transform_publisher_oVeOJWh3irJ86Yi8
/teleop_twist_keyboard
root@ubuntu:~#
```

查看某个节点的信息，输入 ros2 node info 节点名

```
root@ubuntu:~# ros2 node info /originbot_base
/originbot_base
  Subscribers:
    /cmd_vel: geometry_msgs/msg/Twist
    /parameter_events: rcl_interfaces/msg/ParameterEvent
  Publishers:
    /odom: nav_msgs/msg/Odometry
    /originbot_status: originbot_msgs/msg/OriginbotStatus
    /parameter_events: rcl_interfaces/msg/ParameterEvent
    /rosout: rcl_interfaces/msg/Log
    /tf: tf2_msgs/msg/TFMessage
  Service Servers:
    /originbot_base/describe_parameters: rcl_interfaces/srv/DescribeParameters
    /originbot_base/get_parameter_types: rcl_interfaces/srv/GetParameterTypes
    /originbot_base/get_parameters: rcl_interfaces/srv/GetParameters
    /originbot_base/list_parameters: rcl_interfaces/srv/ListParameters
    /originbot_base/set_parameters: rcl_interfaces/srv/SetParameters
    /originbot_base/set_parameters_atomically: rcl_interfaces/srv/SetParametersAtomically
    /originbot_buzzer: originbot_msgs/srv/OriginbotBuzzer
    /originbot_led: originbot_msgs/srv/OriginbotLed
    /originbot_left_pid: originbot_msgs/srv/OriginbotPID
    /originbot_right_pid: originbot_msgs/srv/OriginbotPID
  Service Clients:

  Action Servers:

  Action Clients:

root@ubuntu:~#
```

4）查看话题（topic）

输入：*ros2 topic list*

```
root@ubuntu:~# ros2 topic list
/cmd_vel
/imu
/odom
/originbot_status
/parameter_events
/rosout
/tf
/tf_static
```

查看某个话题（topic）的信息，ros2 topic info 话题名

ros2 topic info /cmd_vel

```
root@ubuntu:~# ros2 topic info /cmd_vel
Type: geometry_msgs/msg/Twist
Publisher count: 1
Subscription count: 1
root@ubuntu:~#
```

查看话题的数据流

ros2 topic echo /cmd_vel

（回到键盘控制小车的窗口，按一下控制键）

```
root@ubuntu:~# ros2 topic echo /cmd_vel
linear:
  x: 0.10294556604732455
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: 0.0
---
linear:
  x: 0.0
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: 0.2058911320946491
---
```

按 ctr+c 终止

5）查看小车的状态

ros2 topic echo /originbot_status

```
root@ubuntu:~# ros2 topic echo /originbot_status
battery_voltage: 11.90999984741211
buzzer_on: false
led_on: true
---
battery_voltage: 11.90999984741211
buzzer_on: false
led_on: true
---
```

电压
蜂鸣器
Led 灯

6）查看数据

话题：/cmd_vel

数据：Twist（标准格式）

from geometry_msgs.msg import Twist

速度：twist.linear.x

转角：twist.angular.z

通过发布话题数据控制小车

ros2 topic pub --rate 1 /cmd_vel geometry_msgs/msg/Twist "linear: x: 0.1, y: 0.0, z: 0.0, angular: x: 0.0, y: 0.0, z: 1.0"

注：这条指令不要直接输入，用 table 键补齐!

```
root@ubuntu:~# ros2 topic pub --rate 1 /cmd_vel geometry_msgs/msg/Twist "linear:
  x: 0.0
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: 0.0" 
```

用左右键，修改 linear.x 和 angular.z 的数值。

```
root@ubuntu:~# ros2 topic pub --rate 1 /cmd_vel geometry_msgs/msg/Twist "linear:
  x: 0.1
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: 1.0"
publisher: beginning loop
publishing #1: geometry_msgs.msg.Twist(linear=geometry_msgs.msg.Vector3(x=0.1, y=0.0, z=0.0), angular=geometry_msgs.msg.Vect
r3(x=0.0, y=0.0, z=1.0))

publishing #2: geometry_msgs.msg.Twist(linear=geometry_msgs.msg.Vector3(x=0.1, y=0.0, z=0.0), angular=geometry_msgs.msg.Vect
r3(x=0.0, y=0.0, z=1.0))
```

按 ctr+c 中断程序。

7）rgt 观察节点（node）和话题（topic）

rqt_graph

